THE EC COUNTEREXAMPLE

Abstract. The main purpose of this note is to give a human exposition of the recent OpenAI counterexample to the C^r, non-invertible case of Shub's entropy conjecture, $r < \infty$. In the final section, we also give some brief, somewhat speculative comments on the invertible case, and in particular on the issue of *degree*. We then analyze the geometry of the OAI example in relation to known local volume growth bounds connected to the entropy conjecture.

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In this note we attempt a deliberately somewhat loose exposition of OpenAI's recent counterexample to the C^1 entropy conjecture for general (i.e. possibly non-invertible) maps.

Theorem A ([Ope26])**.** *For any closed 4-manifold Q and $r < \infty$, there is a C^r map*

$$f : \mathbb{S}^1 \times \mathbb{S}^2 \times Q \to \mathbb{S}^1 \times \mathbb{S}^2 \times Q$$

with $\mathrm{sp}(f_{*,2}) = 2$ *and* $h(f) = 0$.

Though the the OpenAI preprint claims only C^1 regularity, it's easy to see that their construction can be made C^r for arbitrary $r < \infty$ by choosing the parameter L large.

The basic idea of the construction is classical, namely, start with the map $g : z \mapsto z^2$ on $\mathbb{S}^2$ and look for ways to modify it which kill the entropy but not the homology growth. Observe that

$$\Omega(g) = \{0\} \sqcup \mathbb{S}^1 \sqcup \{\infty\},$$

with all entropy carried on the $\mathbb{S}^1$ component, on which g acts as the doubling map. To kill entropy, one needs to destroy this invariant circle, and moreover ensure that no invariant set has non-trivial (positive entropy) dynamics. The canonical illustration of this idea is due to Shub and Sullivan:

Example 0.1. The map $k : z \mapsto 2\dfrac{z^2}{|z|}$ on $\widehat{\mathbb{C}} = \mathbb{S}^2$ satisfies $\deg k = 2$ and $h(k) = 0$. Indeed k is homotopic to $z \mapsto z^2$, but $k^n w \to \infty$ for all $w \neq 0$, so $L(k) = \{0, \infty\}$ is finite and thus $h(k) = 0$.

Importantly, this map is Lipschitz but not C^1, since not differentiable at 0 or ∞. Indeed, Misiurewicz-Przytycki showed that

$$\log \deg(F) \leq h(F)$$

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for any C^1 map from a closed manifold to itself.

The OAI result shows that this theorem cannot be extended to intermediate-degree homology for general maps. Moreover the specific construction used shows that the classical idea *can*, after all, be implemented C^1, *given enough extra space*.

1. The construction of f: heuristic

Again, we start with $g : z \mapsto z^2$ on $\mathbb{S}^2$. Very roughly, the goal is to modify g so that every point not in the basin of 0 eventually lands in the basin of ∞.

For basic topological reasons this is impossible on $\mathbb{S}^2$. To overcome this, first define a C^∞ skew-product map

$$\begin{aligned} G \ : \ \mathbb{S}^1 \times \mathbb{S}^2 &\longrightarrow \mathbb{S}^1 \times \mathbb{S}^2 \\ (t, z) &\longmapsto (t + h(t, z), z^2), \end{aligned}$$

where we identify $\mathbb{S}^1 = \mathbb{R}/N\mathbb{Z}$ with $N \in \mathbb{N}$ large.[1] The smooth translation increment h is usually 1, except near the "gate" coordinate $t_{\text{gate}} = N/2$, where it slows to 0 for z sufficiently small (while remaining 1 for large z).

Formally, write $I_{\text{slow}} = (t_{\text{gate}} - 2, t_{\text{gate}} + 2)$ and $I^-_{\text{slow}} = (t_{\text{gate}} - 2, t_{\text{gate}})$. Then h is characterized by the following properties:

- $h \in C^\infty(\mathbb{S}^1 \times \mathbb{S}^2, [0, 1])$
- $h(t, z) = 1$ if $t \notin I_{\text{slow}}$ or $|z| \geq 3/4$, and $h(t, z) > 0$ provided $t \neq t_{\text{gate}}$
- If $t \in I^-_{\text{slow}}$ and $|z| \leq 1/4$, then $t + h(t, z) < t_{\text{gate}}$.
- Writing $z = re^{i\theta}$, when $(t, z) \in I_{\text{slow}} \times \mathbb{A}_{1/4,3/4}$ we have $\partial_r h(t, z) > 0$.[2]

These properties explain the gate terminology: points with small z-coordinate become trapped on the left side of the gate, while points with z sufficiently large pass through.

Observe that any orbit trapped on the left side of t_{gate} converges to the unique fixed point $(t_{\text{gate}}, 0)$, i.e. if $t \in I^-_{\text{slow}}$ and $|z| \leq 1/4$ then $G^n(t, z) \to (t_{\text{gate}}, 0)$. More generally, define

$$K_0 = \{(t_{\text{gate}}, 0)\}, \qquad \mathcal{T} = \mathbb{S}^1 \times \{|z| = 1\}, \qquad K_\infty = \mathbb{S}^1 \times \{\infty\}.$$

These are all compact, G-invariant sets, and K_∞ is a topological attractor. Moreover

$$\Omega(G) = K_0 \sqcup \mathcal{T} \sqcup K_\infty.$$

On K_∞ we have $G|_{K_\infty} : (t, \infty) \mapsto (t + 1, \infty)$, a rigid rotation, so defining $K := K_0 \cup K_\infty$ we get

$$h(G|_K) = 0.$$

[1] Namely it should be large enough to fit all the intervals defined in §2; obviously the specific choice does not matter.

[2] This assumption will be used only in §3.

On the other hand, clearly $h(G|_{\mathcal{T}}) = \log 2$. Therefore we need to destroy the invariance of $\mathcal{T}$. More generally (simply reiterating the original goal in this setting), we would like to modify G away from K_0 and K_∞ so that every forward orbit not converging to K_0 eventually lands in the basin of K_∞.

How to do this? Consider the forward orbit of a point $x = (t_0, z_0)$ with $t_0 \in [0, 1)$.[3] For $n \in \mathbb{N}$ we write $G^n(t_0, z_0) = (t_n, z_n)$. Then we have the following trivial dichotomy: Either $t_n \leq t_{\text{gate}}$ for all n, or for some n we have $t_n > t_{\text{gate}}$, i.e. the forward orbit of x makes it past the gate. In the first case we have $(t_n, z_n) \to (t_{\text{gate}}, 0)$ as $n \to \infty$, i.e. $\omega_G(x) = K_0$. We will show that we can modify G along the latter orbits to force all of them into the basin of K_∞.

We will start by defining an extremely naive candidate for such a modification, $\widehat{G}$, then see how the definition of f follows naturally from an attempt to make this idea work.

For convenience, write

$$P_{\text{pass}} = \{(t_0, z_0) \in [0, 1) \times \mathbb{S}^2 \ : \ \text{pr}_1 \circ G^n(t_0, z_0) = t_n > t_{\text{gate}} \text{ for some } n \in \mathbb{N}\}.$$

Note that clearly if $(t_0, z_0) \in P_{\text{pass}}$, $|z_n| > 0$ for all n.

Pick some interval $J = [a, a + 1) \subset (t_{\text{gate}} + 2, N - 1)$. Recall that $h \leq 1$, and for all t away from I_{slow} we have $h(t, \bullet) = 1$. It follows that, if the forward orbit of x makes it past the gate, there is a *unique* m such that $t_m \in J$. This m is equal to

$$\tau_J(x) = \min\{n \in \mathbb{N} : \text{pr}_1 \circ G^n(x) \in J\},$$

the *hitting time* of x to J.

For now ignoring continuity issues (which can be dealt with later via bump functions, except where noted), suppose we define $\widehat{G}$ to agree with G away from $J \times \mathbb{S}^2$, while on $J \times \mathbb{S}^2$ we have

$$\widehat{G}(t, z) = (t + 1, z^2 + z^2/|z^2|) = (t + 1, z^2 + e^{2i \arg z}).$$

Though this is not defined when $z = 0$, whenever $t \in J$ and $z \neq 0$ it is easy to see that

$$|\text{pr}_2 \circ \widehat{G}(t, z)| = |z^2 + z^2/|z^2|| > 1.$$

[3]From now on we usually work without comment in the universal cover $\mathbb{R} \times \mathbb{S}^1$ and, abusing notation somewhat, use the same letter to denote both G and its lift.

Also, whenever $x = (t_0, z_0) \in P_{\text{pass}}$ we have $z_n \neq 0$ for all n, thus $\widehat{G}^{\tau_J(x)+1}(x)$ is defined, and

$$\begin{aligned}\widehat{G}^{\tau_J(x)+1}(x) &= \left(t_{\tau_J(x)+1}, z^2_{\tau_J(x)} + \frac{z^2_{\tau_J(x)}}{|z^2_{\tau_J(x)}|}\right) \\ &= \left(t_{\tau_J(x)+1}, z_{\tau_J(x)+1} + \frac{z_{\tau_J(x)+1}}{|z_{\tau_J(x)+1}|}\right).\end{aligned}$$

Since the second coordinate has modulus > 1, and also $|\operatorname{pr}_2 \circ \widehat{G}| \geq |\operatorname{pr}_2 \circ G|$ everywhere, it follows that every $x = (t_0, z_0) \in P_{\text{pass}}$ eventually lands in the basin of K_∞ under $\widehat{G}$, as desired.

Of course it is impossible to make $\widehat{G}$ globally well-defined, let alone C^1. The issue is near $z = 0$. If we could bound $|z_{\tau_J(x)}|$ away from zero, uniformly over all $x \in P_{\text{pass}}$ (equivalently if we could bound τ_J on P_{pass}) we would indeed be done, since the definition of $\widehat{G}$ could be made smooth using appropriate cut-off functions. *However*, it's not hard to see that such a lower bound is impossible: forward orbits from P_{pass} can spend arbitrarily long stretches in I_{slow} and therefore arrive at J with arbitrarily small z-coordinate.[4]

This is where the auxiliary coordinate enters the scene. The first key observation is that, while we cannot uniformly control the z-coordinate along an orbit from P_{pass} *after* it passes the gate, we do have such control *before*, namely $|z_m| > 1/4$ provided $t_m \leq t_{\text{gate}}$. (Otherwise the orbit would never pass the gate, contradicting the definition of P_{pass}.) This implies that along the pre-gate orbit segment, $\arg(z_m)$ *can* be read off by a smooth, globally defined function, namely

$$\alpha : z \mapsto \psi(|z|)\frac{z}{|z|}$$

with $\psi : [0, \infty) \to [0, 1]$ any smooth cutoff with $\psi(s) = 0$ for s sufficiently small and $\psi(s) = 1$ when $s \geq 1/4$.

Roughly speaking the idea is now the following: Pick an interval $I_{I_{\text{store}}} = [b, b+1)$ to the left of t_{gate} with some cushion on either side, say $I_{\text{store}} \subset [3, t_{\text{gate}} - 3]$. For any orbit $G : x_0 \mapsto x_1 \mapsto x_2 \mapsto \cdots$ starting in P_{pass} there is a unique m such that $t_m \in I_{\text{store}}$. At this time, record $\phi(|z_m|)z_m/|z_m|$ in an auxiliary coordinate v. Later when the orbit hits J, use this stored direction data, along with the obvious relation

$$\arg(z_j) = 2^{j-i} \arg(z_i) \forall 0 \leq i < j$$

to achieve the desired push, as in $\widehat{G}$.

[4]For instance, if $y = (t_{\text{gate}}, w)$ with $|w| > 1/4$, then there exists $x \in P_{\text{pass}}$ and $n \in \mathbb{N}$ such that $G^n(x) = y$. But $\lim_{|w| \searrow \frac{1}{4}} \min\{\ell \in \mathbb{N} : \operatorname{pr}_1 \circ G^\ell(y) > t_{\text{gate}} + 1\} = \infty$.

The main problem with this description is that it seems to require squaring the auxiliary coordinate $v \in \mathbb{C}$ at each step between storage and retrieval (since in essence we have stored $\arg(z_m)$ and want to retrieve $\arg(z_{\tau_J(x)+1})$ at the appropriate time). This would be bad for our purposes, since it would produce a new positive entropy factor.

To avoid this, when we pass through the interval I_{store} we want to record $v = \frac{z_{\tau_J(x)+1}}{|z_{\tau_J(x)+1}|}$ directly. As a first approximation, for $t \in I_{\text{store}}$ we could set

$$\overline{v}(t,z) = \left(\psi(z)\frac{z}{|z|}\right)^{2^{\tau_J(t,z)+1}}.$$

Then, indeed, along an orbit segment from $x = (t_0, z_0) \in P_{\text{pass}}$, if $t_m \in I_{\text{store}}$ we get

$$\overline{v}(t_m, z_m) = \left(\frac{z_m}{|z_m|}\right)^{2^{\tau_J(t_m,z_m)+1}} = \left(\frac{z_0}{|z_0|}\right)^{2^{\tau_J(x)+1}} = \frac{z_{\tau_J(x)+1}}{|z_{\tau_J x+1}|},$$

as desired.

Of course, so far this definition simply displaces the entropy/expansion problem, since now unboundedness of τ_J forces unbounded angular derivatives for $\overline{v}$. The first simple yet useful idea is that this can be fixed relatively easily, by introducing a factor of $L^{-\tau_J(t,z)}$ for L very large, and

$$\begin{aligned} v(t,z) &= L^{-\tau_J(t,z)}\left(\psi(z)\frac{z}{|z|}\right)^{2^{\tau_J(t,z)+1}} \\ &= \sum_{n=1}^{\infty} L^{-n}\chi_J(\text{pr}_1 \circ G^n(t,z))\left(\psi(z)\frac{z}{|z|}\right)^{2^{n+1}}. \end{aligned}$$

The equality between the first expression and the infinite series formulation is easy to check. The benefit of the series version is that it "unpacks" the hitting time dependence into more easily analyzed pieces. (We emphasize that, for the series expression especially, we view $J \subset [t_{\text{gate}}, N] \subset \mathbb{R}$ and $G : \mathbb{R} \times \mathbb{S}^2 \to \mathbb{R} \times \mathbb{S}^2$, i.e. this formula takes place on the universal cover. Consequently, for fixed (t, z) at most one term in this infinite series is nonzero.)

Of course, for this (potentially arbitrarily small) stored auxiliary variable to give a definite push later, by this later time it must have definite size. The beauty of the simple weighting-by-L^n idea is that this problem, finally, can easily be dealt with by introducing some trivial dynamics in the auxiliary coordinate, because the modulus of the signal stored at time m is $L^{-(\tau_J(x)-m)}$, i.e. the exponential decay in signal size is a function precisely of the time needed to get from storage to push. *Thus* the stored signal can be restored to definite strength by stipulating the auxiliary dynamics

$v \mapsto Lv$ at intermediate times. Moreover, up to some suitable compactification in the auxiliary coordinate this extra dynamics is benign, since zero entropy.

A last technical (but seemingly unavoidable) point: in order for all of this to work, the auxiliary variable should be empty when orbits arrive at I_{store}. Thus we pick one more unit length interval I_{erase}, with $1 < I_{\text{erase}} < I_{I_{\text{store}}}$. When an orbit from P_{pass} arrives at I_{erase}, its auxiliary variable is set to 0; this guarantees control over this variable for the remainder of the orbit.

Putting everything together, these considerations suggest a candidate map

$$\hat{f} : \mathbb{S}^1 \times \mathbb{S}^2 \times \mathbb{C} \to \mathbb{S}^1 \times \mathbb{S}^2 \times \mathbb{C}$$

by

$$\hat{f}(t, z, w) = (t + h(t, z), z^2 + \chi_J(t)w, C(t)w + \chi_{I_{\text{store}}}(t)v(t, z)),$$

with $C : \mathbb{S}^1 \to \mathbb{R}_{\geq 0}$ any smooth function which is identically zero on I_{erase} and L between I_{store} and J. The proof of the following is essentially contained in the preceding discussion:

Theorem 1.1. $L(\hat{f}) \subset K_0 \sqcup K_\infty$

Naturally, given our use of characteristic functions, the map $\hat{f}$ is not even continuous. Nevertheless it contains all the ideas behind the C^r example. The remaining work (in the next section) largely reduces to book-keeping with bump functions.

2. The construction of f: formal

In this section we modify the candidate map $\hat{f}$ from the preceding section to produce a C^r map f verifying Theorem A. More precisely, we define

$$F : \mathbb{S}^1 \times \mathbb{S}^2 \times \mathbb{C}^2 \to \mathbb{S}^1 \times \mathbb{S}^2 \times \mathbb{C}^2$$

with the desired properties, which is compactly supported in the auxiliary directions and thus obviously realizable inside $\mathbb{S}^1 \times \mathbb{S}^2 \times Q$ for any closed 4-manifold Q.

As before we identify $\mathbb{S}^1 = \mathbb{R}/N\mathbb{Z}$ and work mostly in the universal cover, choosing closed unit length intervals $I_\bullet \subset (0, N)$ with the configuration

$$1 < I_{\text{erase}} < I_{\text{store}} < I_{\text{slow}} = [t_{\text{gate}} - 2, t_{\text{gate}} + 2] < I_{\text{push}} < N.$$

For convenience we also fix $\varepsilon > 0$ small enough that the ε-neighborhoods of these intervals are all pairwise disjoint. (Again, $t_{\text{gate}} = N/2$, and N is simply chosen large enough to make all of this fit.)

We will also use the following C^∞ cutoffs and bump functions:

- $\eta = \eta_{\text{erase}} : \mathbb{S}^1 \to \mathbb{R}$ such that $\eta|_{I_{\text{erase}}} \equiv 0$ and $\eta|_{\mathbb{S}^1 \setminus I_{\text{erase}}} \equiv 1$.

- $\phi_{\text{ann}} : \mathbb{S}^2 = \widehat{\mathbb{C}} \to [0,1]$ which is identifically 1 on $\mathbb{A}_{1/R,R}$ and supported in $\mathbb{A}_{1/2R,2R}$, for some $R > 4$;
- $\phi_{\text{store}} : \mathbb{R} \to [0,1]$ with $\phi_{\text{store}}|_{I_{\text{store}}} \equiv 1$ and $\text{supp}(\phi_{\text{store}}) \subset N_\varepsilon(I_{\text{store}})$;
- $\beta_1, \beta_2 : \mathbb{R} \to [0,1]$ with $\text{supp}(\beta_i) \subset N_\varepsilon(I_{\text{push}})$, $|\text{supp}(\beta_i)| < 1$, and $\max\{\beta_1(s), \beta_2(s)\} = 1$ for all $s \in I_{\text{push}}$.
- $c : \mathbb{C} \to \overline{\mathbb{D}}_2^2$ is given by $c(z) = p(|z|)z$ for $p : [0,\infty) \to [0,2]$ smooth, satisfying
  - $p(r) = r$ for all $0 \le r \le 1$,
  - $p(r)$ is nondecreasing on $[0, 2L]$, and
  - $p(r) = 0$ for all $r \gg 0$.

Finally we let $L \gg 2$ denote a for-now-unspecified constant, eventually taken as large as needed to make the argument work.

Remark 1. The definition of the β_i is chosen so that, for every $x = (t_0, z_0) \in P_{\text{pass}}$ there is some $n \in \mathbb{N}$ and $i \in \{1,2\}$ such that $\beta_i(t_n) = 1$, and on the other hand, for each fixed $i \in \{1,2\}$ there is *at most* one n such that $\beta_i(t_n) > 0$. (Here as before we write $(t_n, z_n) = G^n(t_0, z_0)$ when the starting point is clear.) By ensuring unique hitting times to $\text{supp}(\beta_i)$ one avoids possible cancellation between terms with different phases in the series definition for σ_i (cf. hitting times series expansion above). By requiring that some $\beta_i(t_m) = 1$ for some m, one ensures that at least one of the stored auxiliary signals is strong enough to produce the required push. Our desire for both these properties explains the use of two bump functions in this region rather than one, and correlatively our switch to $\mathbb{C}^2$ (rather than $\mathbb{C}$) for the auxiliary variables, but it seems plausible that one could reduce the dimension of the auxiliary manifold by at least 1 with a little more care or a slightly different technique.

Remark 2. The cut-off function c is necessary to make the required dynamics on the auxiliary factor $\mathbb{C}^2$ realizable on the compactification $(\mathbb{S}^2)^2$ (or any compactification). More precisely, as indicated above, one must be able to interpolate continuously between the auxiliary dynamics $\mathbf{w} \mapsto L\mathbf{w}$ on the region $I_{\text{store}} \le t \le I_{\text{push}}$, and the reset $\mathbf{w} \mapsto \mathbf{0}$ when $t \in I_{\text{erase}}$. While the multiplication-by-L dynamics is defined globally on $(\mathbb{S}^2)^2$ and unproblematic from an entropy perspective (the non-wandering set consists of 4 fixed points), without some cut-off it has degree 1 and thus no hope of interpolating continuously to a constant map. On the other hand, this expansion is needed only to amplify exponentially small signals to norm 1, so the cut-off c preserves it on the relevant auxiliary region.

Now for $i = 1, 2$ define

$$\sigma_i : \mathbb{R} \times \mathbb{S}^2 \longrightarrow \mathbb{C}$$

by

$$\sigma_i(t,z) = \phi_{\text{store}}(t) \sum_{n=1}^{\infty} L^{-n} \beta_i(\text{pr}_1 \circ G^n(t,z)) \left(\phi_{\text{ann}}(z) \frac{z}{|z|} \right)^{2^{n+1}}$$

and $\sigma(t,z) = (\sigma_1(t,z), \sigma_2(t,z))$. Similarly for convenience we write

$$\beta(t) = (\beta_1(t), \beta_2(t)) \in \mathbb{R}^2, \quad \mathbf{c}(\mathbf{w}) = (c(w_1), c(w_2)) \in \mathbb{C}^2.$$

Then, finally,

$$F : \mathbb{S}^1 \times \mathbb{S}^2 \times \mathbb{C}^2 \to \mathbb{S}^1 \times \mathbb{S}^2 \times \mathbb{C}^2$$

is defined by

$$\boxed{F : \begin{pmatrix} t \\ z \\ \mathbf{w} \end{pmatrix} \longmapsto \begin{pmatrix} t + h(t,z) \\ z^2 + L^2 \langle \mathbf{c}(\mathbf{w}), \beta(t) \rangle \\ L\eta(t)\mathbf{c}(\mathbf{w}) + \sigma(t,z) \end{pmatrix}.} \tag{1}$$

Since the cut-off function $c : \mathbb{C} \to \overline{\mathbb{D}}^2_2$ vanishes near ∞ it obviously extends to $\widehat{\mathbb{C}}$, thus F extends to

$$f : \mathbb{S}^1 \times \mathbb{S}^2 \times (\mathbb{S}^2)^2 \to \mathbb{S}^1 \times \mathbb{S}^2 \times (\mathbb{S}^2)^2.$$

We emphasize again that there is nothing special about the choice of auxiliary factor $\mathbb{S}^2 \times \mathbb{S}^2$. Indeed any closed 4-manifold Q will do: simply pick a smooth chart $\mathbb{C}^2 \cong U \subset Q$, transport F to $\mathbb{S}^1 \times \mathbb{S}^2 \times U$ via this chart, and observe that $\mathbf{c}$ vanishes near ∂U, thus extends by zero to Q just as above.

For concreteness, however, we focus on f. Write $M = \mathbb{S}^1 \times \mathbb{S}^2 \times (\mathbb{S}^2)^2$. The following properties of F and/or f are in fact obvious.

- **$\mathbf{C^r}$-regularity**: Every function involved in defining F is C^∞ except for σ. Moreover for every finite r, the C^r-norm of the terms in the series defining σ grows at most exponentially in n, thus by choosing L sufficiently large (depending on r) we can ensure $\sigma \in C^r$, for any $r < \infty$. *Therefore F can be made any finite regularity.* (It *cannot* be made C^∞: while the C^r-norms of the series terms grow exponentially, the exponential growth rate goes to ∞ with r.)

- **Homology**: If $\alpha \in H_2(M, \mathbb{R})$ denotes the class represented by $\{t\} \times \mathbb{S}^2 \times \{\mathbf{w}\}$ for any $t \in \mathbb{R}$ and $\mathbf{w} \in (\mathbb{S}^2)^2$, then $f_* \alpha = 2\alpha$. Thus $\text{sp}(f_{*,2}) \geq 2$.

  Indeed, f is homotopic to the map $(t, z, \mathbf{w}) \mapsto (t, z^2, \mathbf{0})$, which acts by the identity on $H_1(M, \mathbb{R})$, multiplication by 2 on H_2 and H_3, and the zero map on H_k, $4 \leq k \leq 7$.

Thus Theorem A reduces to the following claim.

Proposition 2.1. *Define*

$$Z_0 = K_0 \times (\mathbb{S}^2)^2 \text{ and } Z_\infty = K_\infty \times \{\mathbf{0}\}.$$

Then Z_0 and Z_∞ are compact, f-invariant, and satisfy $h(f|_{Z_0}) = h(f|_{Z_\infty}) = 0$. Moreover we have

$$\omega_f(x) \subset Z_0 \sqcup Z_\infty$$

for all $x \in M$.

Corollary 2.2. $h(f) = 0$.

Proof. By the proposition, $L(f) \subset Z := Z_0 \cup Z_\infty$. Also clearly $0 = h(f|_Z) \geq h(f|_{L(f)})$, so by the variational principle $h(f) = h(f|_{L(f)}) = 0$. □

The compact invariance and zero entropy claims in Proposition 2.1 are trivial. (For instance, $h(f|_{Z_0})$ follows from the fact that the bump function p defining c is monotonic on its image $[0, 2]$.) As in the preceding section, the proof that

$$L(f) \subset Z_0 \cup Z_\infty$$

reduces to showing that the forward orbit of any $x = (t_0, z_0, \mathbf{w}_0)$ with $t_0 \in [0, 1)$ which makes it past the gate gets pushed into the basin of Z_∞ on I_{push}. This is a simple calculation, essentially identical to that from the preceding section.

3. Comments

We conclude with some additional comments on this construction. First we note some possible directions for, and obstructions to, generalizing to the diffeomorphism case, in particular related to additional topological constructions which appear for mappings of non-zero degree. Then we study this example from the perspective of volume growth of submanifolds, and particular start to describe the mechanism through which it produces *stable local volume growth*.

Given the obvious non-invertibility of f, one naturally wonders whether the construction used here can be adapted to produce an invertible example. This is especially interesting given the limited class of invertible entropy conjecture counterexamples even in the C^0 category. The canonical existing example *is* essentially an invertible adaptation of the Shub-Sullivan idea. Namely, whereas the Shub-Sullivan example can be viewed as the suspension of the doubling map on the circle composed with a north-south dynamic in the vertical direction (so that everything away from the poles is wandering), in the invertible adaptation one takes the suspension of a hyperbolic toral automorphism, again composed with a north-south map. The only problem is that the resulting space is not a manifold (neighborhoods of the poles

are cones over tori). The remaining work uses some PL-embeddings and gluings to resolve this issue. See [Pug75] for details.

By analogy with the above construction, one might try to start with a C^∞ diffeomorphism homotopic (in some controlled way) to this homeomorphism counterexample, then couple with some (invertible, zero entropy, finite-regularity) auxiliary dynamics to recover the wandering behavior of the C^0 counterexample and thus force entropy zero.

Needless to say, it is far from obvious how to implement this idea. In particular we want to emphasize that the success of this coupling strategy in producing the OAI example seems to depend quite strongly on the non-invertibility of the auxiliary dynamics. Even if some coarsely similar coupling strategy worked in the invertible setting, the details would necessarily look very different.

3.1. **Degree.** We observed above that the map $f : M \to M$ is homotopic to the map $(t, z, \mathbf{w}) \mapsto (t, z^2, \mathbf{0})$. In particular, $\deg(f) = 0$.

When f is C^1 and $|\deg(f)| \geq 2$, meanwhile, Misiurewicz and Przytycki's result implies that $h(f) > 0$. It is then natural to ask about the remaining possibility.

Question 3.1. *Does there exist a closed manifold N and $f : N \to N$ C^1 with $\deg(f) = 1$, $\mathrm{sp}(f_{*,k}) > 1$ for some $1 \leq k \leq \dim(N) - 1$, and $h(f) = 0$?*

We emphasize that f here is still allowed to be non-invertible. Thus such an example would provide an intermediate step to producing a diffeomorphism EC counterexample.

On the other hand, even such an example seems out of reach for the construction used above. The issue is that the orbit-tracking argument strongly depends on resetting all auxiliary variables to 0 in the interval I_{erase}. To accommodate this, we are forced to collapse all of M into a tubular neighborhood of $\mathbb{S}^1 \times \mathbb{S}^2 \times \{0\}$ under f, which immediately implies degree 0 (and more). The issue here is more serious, topologically, than mere non-invertibility: the auxiliary coordinate strategy, at least as realized here, appears to force $\ker(f_*)$ large.

More generally, for degree 1 maps Poincaré duality imposes additional constraints which do not hold in the degree 0 case.

3.2. **Local volume growth.** Yomdin's proof of the entropy conjecture for C^∞ maps begins with the estimate

$$\log \mathrm{sp}(f_{*,k}) \leq h(f) + \mathrm{LV}_k(f),$$

where $\mathrm{LV}_k(f)$ denotes the maximum *local volume growth* of smooth k-disks in M under f, i.e.

$$\begin{aligned}\mathrm{LV}_k(f) &= \sup_{\psi\in C^\infty(\mathbb{D}^k,M)} \mathrm{LV}(f,\psi)\\ &= \sup_{\psi\in C^\infty(\mathbb{D}^k,M)} \lim_{\varepsilon\to 0}\limsup_{n\to\infty}\frac{1}{n}\log^+\left(\sup_{x\in M}\mathrm{vol}\left(f^n\circ\psi|_{\psi^{-1}(B_f(x,n,\varepsilon))}\right)\right).\end{aligned}$$

He then shows (in the real substance of the proof) that $\mathrm{LV}(f)=\max_{1\le k\le\dim(M)}\mathrm{LV}_k(f)=0$ for C^∞ maps [Yom87].

While examples show that $\mathrm{LV}(f)$ can be positive in any finite regularity, an entropy conjecture counterexample implies more, namely we must have local volume growth which is *stable* over some submanifold's entire homotopy class.[5]

More precisely, for a closed k-manifold S and a C^1 map $\psi:S\to M$, write $\widehat{\psi}\sim\psi$ to mean that $\widehat{\psi}\in C^1(S,M)$ and $\widehat{\psi}$ is homotopic to ψ. Then we define the *homotopy local volume growth* of ψ under f by

$$\mathrm{HLV}(f,\psi)=\lim_{\varepsilon\to 0}\liminf_{n\to\infty}\frac{1}{n}\log\left(\inf_{\widehat{\psi}\sim\psi}\sup_{x\in M}\mathrm{vol}\left(f^n\circ\widehat{\psi}|_{\widehat{\psi}^{-1}(B_f(x,n,\varepsilon))}\right)\right).$$

Now suppose ψ represents a k-homology class α (i.e. $\psi_*[S]=\alpha$) with $\|f_*^n\alpha\|\sim\lambda^n$, but $h(f)=0$. Then we must have

$$\mathrm{HLV}(f,\psi)\ge\log\lambda.$$

This is precisely the situation for the f defined here, where now

$$\begin{aligned}\psi:\mathbb{S}^2&\to\mathbb{S}^1\times\mathbb{S}^2\times(\mathbb{S}^2)^2\\ z&\mapsto(0,z,\mathbf{0}).\end{aligned}$$

Indeed $h(f)=0$ and $f_*\psi_*[\mathbb{S}^2]=2\psi_*[\mathbb{S}^2]$. Thus

$$\mathrm{HLV}(f,\psi)\ge\log 2.$$

In particular we must have local volume growth, uniform in n, in all of the main sphere fibers

$$S_{t,\mathbf{w}}:=\{t\}\times\mathbb{S}^2\times\{\mathbf{w}\},\qquad (t,\mathbf{w})\in\mathbb{S}^1\times(\mathbb{S}^2)^2.$$

More precisely, fix $1<\lambda<2$. Then for all $\varepsilon>0$ there exists $N_\varepsilon\in\mathbb{N}$ such that for all $n\ge N_\varepsilon$ and $(t,\mathbf{w})\in\mathbb{S}^1\times(\mathbb{S}^2)^2$, there exists $x\in S_{t,\mathbf{w}}$ such that

$$\lambda^n<\mathrm{vol}(f^n(S_{t,\mathbf{w}}\cap B_f(x,n,\varepsilon))). \tag{2}$$

Using the canonical identification $S_{t,\mathbf{w}}=\mathbb{S}^2$, one would like to understand the distribution of the set of $x\in\mathbb{S}^2$ verifying (2), as a function of n,ε,t, and $\mathbf{w}$. Put simply:

[5]For further discussion of this point and the definitions used here, see [Kod26].

Question 3.2. *Where in each main sphere fiber does local volume growth occur, and by what mechanism is it produced?*

The remainder of this section will be devoted to (at least partially and somewhat sketchily) answering this question.

For context, note that the local volume growth for the standard Lipschitz Shub-Sullivan example $z \mapsto 2z^2/|z|$ is concentrated at the repelling fixed point 0. By slowing down the radial attraction at the attracting fixed point ∞ one can get local volume growth there as well, and in fact (a quick calculation shows) at every point in $\mathbb{S}^2$.[6] In both cases, however, the mechanism through which local volume growth is produced is clear, namely it is produced (in this case) by fixed points with exponential angular expansion and insufficient compensatory radial attraction. A map cannot, of course, be differentiable at points with this property.

The standard Yomdin-Margulis example of f C^r with $\mathrm{LV}(f) > 0$ can be built locally as follows. We start with a hyperbolic linear map

$$A = \begin{bmatrix} 1/\lambda & 0 \\ 0 & \lambda \end{bmatrix} : \mathbb{R}^2 \to \mathbb{R}^2$$

with $\lambda > 1$. Pick $0 \neq p_0 = (x_0, 0) \in W^s_A(0) = \mathbb{R} \times \{0\}$. Pick a neighborhood $U \ni p$ such that $A(U) \cap U = \varnothing$ and a slightly smaller neighborhood $V \ni p$ with $\overline{V} \subset U$. Pick a bump function φ supported on U with $\varphi|_V \equiv 1$. Finally define $f : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$f := A + \varphi \cdot g,$$

where

$$g(x, y) := \left(0, (x - x_0)^{2r+1} \sin\left(\frac{1}{x - x_0}\right)\right).$$

It is then easy to check that for any $\varepsilon > 0$, if $\gamma = [x_0, x_0 + \varepsilon) \times \{0\}$ then

$$\text{length } f^n(\gamma \cap B_f(p_0, n, \varepsilon)) \sim \lambda^{n/2r}.$$

The idea is that f first maps γ to an infinitely oscillatory curve, whose branches subsequently expand in the unstable direction of 0 while being pinned in place along its stable manifold.

We now turn to the analysis of f as defined in §2. First observe that any forward f-orbit which does not visit the region

$$\mathcal{B} := N_\varepsilon(I_{\text{store}}) \times \mathbb{A}_{1/2R,2R} \times (\mathbb{S}^2)^2$$

[6]The calculation in question: Show that for the map $\alpha : \mathbb{R}^2 \to \mathbb{R}^2$ given in polar coordinates by $(r, \theta) \mapsto (\lambda^{-1} r, 2\theta)$, for any $\varepsilon > 0$ and any open set $U \subset \mathbb{D}^2_\varepsilon$ there is a $C > 0$ such that

$$\text{area}(\alpha^n(U)) \geq C(2\lambda^{-2})^n,$$

where as usual area is calculated with multiplicity. Now take $\lambda \searrow 1$.

agrees with a forward orbit for the map

$$\mathfrak{f} : (t, z, \mathbf{w}) \mapsto (t + h(t, z), z^2 + L^2\langle \mathbf{c}(\mathbf{w}), \beta(t)\rangle, L\eta(t)\mathbf{cw}),$$

since σ vanishes everywhere along this orbit. Moreover, if an orbit stays distance ε away from $\mathcal{B}$, its Bowen balls $B_f(x, n, \delta)$ agree with those of $\mathfrak{f}$, as do the dynamics up to time n. This latter map $\mathfrak{f}$, on the other hand, is C^∞, and thus has zero local volume growth by Yomdin.

This in mind, we will first sketch the answer to Question 3.2 in the specific case

$$S := \mathbb{S}^2_{t_0,\mathbf{0}}$$

for fixed $t_0 \in I_{\text{store}}$. We also write

$$\widehat{S}_t = \{t\} \times \mathbb{S}^2 \subset \mathbb{S}^1 \times \mathbb{S}^2.$$

Let

$$n_0 = n_0(t_0) = \min\{n \in \mathbb{N} : G^n(t_0, z) = (t_{\text{gate}}, w) \text{ for some } z \in \mathbb{S}^2 \text{ and } |w| \le 1/4\}.$$

By the last property in the definition of h we have

$$G^{n_0}(\widehat{S}_t) \pitchfork \widehat{S}_{t_{\text{gate}}},$$

and indeed it is easy to see that

$$G^{n_0}(\widehat{S}_t) \cap \widehat{S}_{t_{\text{gate}}} = \{|z| = r\} =: \mathbb{S}^1_r \subset \widehat{S}_{t_{\text{gate}}}$$

for some $r \le 1/4$. Moreover, we have

$$G^{-n_0}(\mathbb{S}^1_r) = \mathbb{S}^1_{r_0} \subset \widehat{S}_{t_0}$$

for some $1/4 < r_0 < 1$.

Now fix $i \in \{1, 2\}$. The map

$$\sigma_i(t_0, \bullet) : S \cong \widehat{\mathbb{C}} \to \mathbb{C}$$

has the following structure: $\sigma_i(t_0, z) = 0$ if $|z| \le r_0$, and the circle $\{|z| = r_0\}$ is accumulated by annuli

$$A_k = \{a_k \le |z| \le b_k\},$$

defined for all k larger than some N_0 with $a_{k+1} < b_{k+1} < a_k < b_k$ and $a_k \searrow r_0$, such that for all $k \ge N_0$,

- $\sigma_i|_{\partial A_k} \equiv 0$
- $\sigma_i > 0$ on $\overset{\circ}{A}_k$
- $\max\{|\sigma_i(t_0, z)| \ : \ z \in A_k\} = L^{-k}$, with this supremum attained everywhere on some sub-annulus

$$A'_k = \{a'_k \le |z| \le b'_k\}, \quad a_k < a'_k < b'_k < b_k.$$

- σ_i maps every core circle $\{|z| = c\}$ $(a_k < c < b_k)$ to some other circle $\{|z| = c'\}$ $(0 < c' \le L^{-k})$ with degree 2^{k+1}, i.e. σ_i wraps every core circle of A_k 2^{k+1} times around the origin.

An immediate consequence of these properties is that σ_i maps A_k onto $\overline{\mathbb{D}}^2_{L^{-k}}$ with multiplicity 2^{k+2}, thus

$$\int_{A_k} \left\|\Lambda^2 D\sigma_i|_{A_k}\right\| d\,\mathrm{Leb} =: \mathrm{vol}(\sigma_i|_{A_k}) = 2^{k+2} \cdot \pi(L^{-k})^2.$$

Now let $x_0 = (t_{\mathrm{gate}}, 0, \mathbf{0}) \in Z_0$, so $f(x_0) = x_0$. Fix $\varepsilon > 0$. We have

Claim 3.3. *For some uniform $n_1 \in \mathbb{N}$, $n_1 \ge n_0$, we have $f^{n_1}A_k \subset B_\varepsilon(x_0)$ for all $k > N_0$.*[7]

Proof. Pick some $n' \in \mathbb{N}$ such that $(1/4)^{2^{n'}} < \varepsilon$ and set $n_1 = n_0 + n'$. □

Recall that the subscript of A_k is precisely equal to its hitting time to $\mathrm{supp}(\beta_i)$. Since the translation increment h is bounded from below on $[0, N] \setminus (t_{\mathrm{gate}} - \varepsilon, t_{\mathrm{gate}} + \varepsilon)$ we get

Claim 3.4. *For some uniform $n_2 \in \mathbb{N}$ we have $\mathrm{pr}_1 \circ f^n(A_k) \subset (t_{\mathrm{gate}}, t_{\mathrm{gate}} + \varepsilon)$ provided $k \ge n + n_1 + n_2$.*

Finally we choose $n_3 \in \mathbb{N}$ large enough that $L^{-n_3} < \varepsilon$. It follows that

$$f^n(A_k) \subset B_\varepsilon(x_0)$$

provided $n_1 \le n \le k - n_1 - n_2 - n_3$, or, rearranging, that

$$f^{n_1}(A_k) \subset B_f(x_0, n, \varepsilon)$$

provided $k \ge n + n_1 + n_2 + n_3$. Then

$$\begin{aligned}
\mathrm{vol}\, f^n(S \cap f^{-n_1} B_f(x_0, n, \varepsilon)) &\ge \sum_{k \ge n+n_1+n_2+n_3} \mathrm{vol}\, f^n(A_k) \\
&\ge \mathrm{vol}\, f^n(A_{n+n_1+n_2+n_3}) \\
&= L^{2(n-1)}\, \mathrm{vol}(\sigma_i|_{A_{n+n_1+n_2+n_3}}) \\
&= \pi L^{2(n-1)} \cdot 2^{n+n_1+n_2+n_3+2} \cdot L^{-2(n+n_1+n_2+n_3)} \\
&= \frac{\pi 2^{2+n_1+n_2+n_3}}{L^{2+n_1+n_2+n_3}} 2^n = C(n_1, n_2, n_3) \cdot 2^n.
\end{aligned}$$

[7]For convenience we use the ℓ^∞ metric on the product manifold M induced by the standard metrics on its factors, so $B_\varepsilon(x_0) = (t_{\mathrm{gate}} - \varepsilon, t_{\mathrm{gate}} + \varepsilon) \times (\mathbb{D}^2_\varepsilon)^3$.

Fixing an (n_1, ε)-cover of S, with cardinality $r_f^S(n_1, \varepsilon)$, this calculation moreover allows us to localize the local volume growth to $A_{n+n_1+n_2+n_3}$: for all $n > n_1$ there exists $z \in A_{n+n_1+n_2+n_3}$ such that

$$\operatorname{vol} f^n(A_{n+n_1+n_2+n_3} \cap B_f(z, n, \varepsilon)) \geq \frac{C(n_1, n_2, n_3)}{r_f^S(n_1, \varepsilon)} \cdot 2^n.$$

Since the coefficient here depends only on ε, this gives the beginnings of an answer to Question 3.2, at least on a sphere fiber $S = \mathbb{S}^2_{t,\mathbf{0}}$ for $t \in I_{\text{store}}$. The generalization to more general sphere fibers (or more generally spheres homotopic to such fibers), however, should not be difficult: the essential observation is simply that any such fiber contains a piece which passes through $I_{I_{\text{store}}}$ and then intersects $\{(t_{\text{gate}}, z, \mathbf{w}) : |z| \leq 1/4, \mathbf{w} \in (\mathbb{S}^2)^2\}$ transversally, *in uniformly bounded time*. Then the argument can be repeated as before.

As far as *mechanism* is concerned, the description above in a sense shows that local volume growth for f follows from a combination of the Shub-Sullivan and Margulis-Yomdin examples. Here the function σ_i produces infinitely oscillatory behavior in the auxiliary, *unstable* direction, which is held in place near the fixed point x_0 for a long time by the stable (indeed super-attracting) dynamics in the z-coordinate and slow saddle-node behavior in the t-coordinate. However, the oscillation in $|\sigma_i|$ alone is not sufficient to produce local volume growth. For this, one needs exponential angular growth accumulating on the circle $\{|z| = r_0\}$ (which is, indeed, the boundary of $W^s(x_0) \cap S$). This is contributed by the $\left(\frac{z}{|z|}\right)^{2^n+1}$ factor, giving the result.

References

[Kod26] A. Kodat. *Some results on intersections and entropy*. PhD thesis, CUNY Graduate Center, 2026. https://axelkodat.github.io/thesis.pdf.

[Ope26] OpenAI. A C^1 Counterexample to the Entropy Conjecture. OpenAI Math Release preprint. https://github.com/openai/math/blob/main/preprints/A-C1-Counterexample-to-the-Entropy-Conjecture-September-25-2026/article.pdf, 2026.

[Pug75] C. Pugh. On the entropy conjecture: a report on conversations among R. Bowen, M. Hirsch, A. Manning, C. Pugh, B. Sanderson, M. Shub and R. Williams. In A. Manning, editor, *Dynamical Systems—Warwick 1974*, Lecture Notes in Mathematics, pages 257–261. Springer Berlin Heidelberg, Berlin, Heidelberg, 1975.

[Yom87] Y. Yomdin. Volume growth and entropy. *Israel Journal of Mathematics*, 57(3):285–300, 1987.